

4.12.2 Writing functional programs

4.12.2.1 Functional language programs

Content	Additional information
Show experience of constructing simple programs in a functional programming language.	The following is a list of functional programming languages that could be used: • Haskell • Standard ML • Scheme • Lisp. Other languages with built-in support for programming in a functional paradigm as well as other paradigms are: • Python • F# • C# • Scala • Java 8 • Delphi XE versions onwards • VB.NET 2008 onwards.
Higher-order functions.	A function is higher-order if it takes a function as an argument or returns a function as a result, or does both.

Content	Additional information
Have experience of using the following in a functional programming language: • <code>map</code> • <code>filter</code> • <code>reduce</code> or <code>fold</code>.	<i>map</i> is the name of a higher-order function that applies a given function to each element of a list, returning a list of results. <i>filter</i> is the name of a higher-order function that processes a data structure, typically a list, in some order to produce a new data structure containing exactly those elements of the original data structure that match a given condition. <i>reduce</i> or <i>fold</i> is the name of a higher-order function which reduces a list of values to a single value by repeatedly applying a combining function to the list values.